
Can coordination improve energy efficiency in a LEO handover simulator?

Critical literature review

8 September 2026 • Review of the r6.1 project package • Primary-source evidence through the review date

Executive assessment

A positive EE marginal is plausible under rate-target power control, but neither its existence nor its size is established for the successor. With less airtime per user, sustaining a rate target requires a higher spectral efficiency and generally higher transmit power. Association also changes interference, feasibility and active hardware. These mechanisms create opportunities that the old occupancy-insensitive RF rule suppressed. They do not make balancing universally preferable: activation costs favor consolidation, while congestion, poor channels and interference can favor spreading.

The **best prior is a coordinator that scores complete association configurations and commits the selected configuration atomically**, using the same information and whole-network objective as the baseline. It should represent rate-target power, co-channel interference, active-resource costs and service constraints, permitting transfers, swaps and evacuations. This is an inference from satellite resource-allocation evidence, coordination-graph results and optimization-assisted learning—not a demonstrated winning architecture for this project. A third independent per-user scalar, added to two existing heads and decoded by separate argmax operations, has a weaker prior when benefits require simultaneous adoption. [Wang et al. DCG DROO](#)

The literature distinguishes **coordination value, learning value and implementation speed**. Beating two learned heads can reflect correction of their approximation errors; matching a nominal optimizer faster establishes computational value, not additional EE. No reviewed source establishes that all three heads must have positive conditional marginals, or that learning must beat an equal-information optimizer.

The package already reports positive restricted search results, so it does not demonstrate a universal absence of coordination headroom. Conversely, its later approximately 2.9% result does not identify a renewal mechanism or predict successor performance. These are supplied, unreplicated project findings. No primary source verifies that attribution, an expected successor effect size, or a universally sufficient seed count for detecting 1–3%.

Scope, evidence and reading rules

This review read files 00, 01, 02, 11, 16 and 20, together with the result, training, service, evaluation and differential-audit records needed to reconcile them. **File 20 controls the successor specification.** It supersedes file 11's fixed-RF priority and resolves file 16's concern that a constant SINR target had been substituted for a rate target. The primary is memoryless nominal per-user rate-target control with equal-airtime, full-band TDM, discrete ACM and a cap; fixed RF plus ACM is the reference. No simulator experiments were performed and no parameter values are proposed here.

Evidence labels have strict meanings. **S** means the named primary source supports the specified, bounded claim; it does not prove project causation. **A** means a mathematical or logical derivation in this review, explicitly distinguished from an empirical finding. **P/U** means a fact reported in the supplied package, whose implementation or causal contribution cannot be independently verified from an external primary source. **U** marks an unresolved or unavailable primary-source claim. This labeling applies to every project-specific statement below, including the hypothesis tables.

The searches covered satellite load, power and beam hopping; wireless MARL and handover; fractional objectives; simulator calibration; and dependent, small-sample inference. Journal/conference originals, standards and author manuscripts were prioritized. ResearchGate links below provide copies of the original papers; upload provenance was not independently authenticated. This is a critical review, not a registered systematic review or proof that no unlocated paper exists. Most research sources are from 2015–2026; earlier standards and foundational studies are named explicitly. The annotations state the exact supported claim and its limit. Cross-domain results are identified as transfers, not satellite evidence.

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1. When user load changes satellite energy

1.1 Rate targets restore a mechanism, with important qualifications

A — ideal link derivation. For user u scheduled for fraction τ of a full bandwidth W , let h be the power gain and I the interference power during its transmission. Under a Shannon reference link,

$$R = \tau W \log_2(1 + hp/(N_0W + I))$$

$$p_{\text{required}} = ((N_0W + I)/h) \times (2^{r_{\text{target}}/(\tau W)} - 1).$$

With equal airtime $\tau = 1/n$, occupancy n enters the exponent. The beam's average PA supply power is the airtime-weighted sum of the PA supply function evaluated at each scheduled user's RF power. It is not generally the PA function of the maximum user power, nor the sum of all users' simultaneous powers. For FDM, each user's occupied bandwidth changes the noise bandwidth and interference overlap as well as its rate; replacing W by a reduced bandwidth only in one term is inconsistent. This derivation is an idealized explanatory limit, not a proposed modem or calibrated payload model.

A — successor implication. File 20 replaces continuous inversion with the lowest ACM mode that can meet the nominal rate under the assigned airtime, followed by capped power inversion. Thus load-dependent power is stepwise: occupancy changes that do not cross a mode boundary need not change RF power. A cap can create flat power regions and missed targets. The specified infeasible attempt remains in the transmission, interference and energy accounting, with actual partial delivery; it is not silently removed or repacked. The target is a control target, not a finite-demand cap on scored bits. [P20; P/U]

The DVB-S2 standard supplies finite modulation/coding choices and reference decoding thresholds. Its Table 13 uses spectral efficiency per symbol and specified reference conditions; bandwidth conversion, waveform overhead and implementation losses still require explicit modeling. It does not certify Shannon capacity as achieved throughput or validate the project's thresholds. ITU-R S.2131 explicitly treats ACM availability and throughput statistically: a nominal or median link budget is not an expected delivered-rate calculation. [DVB-S2 ITU-R S.2131](#)

A — balancing is conditional. With identical channels, fixed interference, continuously feasible equal rate targets and a fixed set of active beams, the ideal RF requirement is convex in occupancy. Distributing users can lower variable RF demand. Unequal channels, discrete modes, caps, interference and nonlinear PA supply conversion remove any general occupancy-only dominance. Adding an active RF chain can cost more than the RF reduction. Constant target SINR, by contrast, does not itself make occupancy raise required power: it can simply let per-user rates fall as airtime shrinks.

1.2 Five distinct mechanisms must not be conflated

Mechanism	What changes with association or load?	Energy consequence and evidence ceiling
Rate-target control	Airtime/bandwidth requirement and link gain change required RF.	A variable-power opportunity exists under feasible targets. It is not an empirical guarantee for file 20; the derivation above is A.
Co-channel interference	The selected transmitters, powers and simultaneous schedules change each other's required power or achieved rate.	Joint allocation can reduce a coupled objective. Energy accounting must use the same radiation for wanted signal, interference and PA draw. Satellite examples: Wang , Liao-Kaneko .
Sleeping / beam hopping	Active RF resources and illumination duty cycle change.	Savings depend on switchable hardware and standby/fixed floors. Fewer lit beams need not mean fewer powered chains on a shared payload. Fastenbauer , Cocco .
PA operating point / back-off	RF output, efficiency and possibly multicarrier distortion change.	RF joules and payload DC joules are different objectives. A scalar efficiency curve alone does not model distortion. DVB implementation guide , Cocco .
Flexible power sharing	Power/bandwidth can move among beams under shared amplifier and payload limits.	Flexibility can improve demand matching without reducing total power. Fixed-budget throughput gains are not isolated load-to-energy savings. Cocco .

The claim that satellite downlinks must use fixed RF is too strong. The DVB-S2 implementation guide expressly accommodates power control and ACM. That establishes compatibility, not that a particular LEO payload implements the successor's per-user law. [DVB implementation guide, §4.1.2](#)

1.3 What the quantitative satellite papers actually show

Primary study	Reported comparison and result	Exact supported claim; excluded interpretation
Wang et al., 2025/2026	Fig. 6 reports 25.54% lower average power than its no-load-balancing benchmark at the paper's 20-Gbps load point.	A load-balancing DROP is available in multi-satellite beam hopping. Its objective counts downlink RF energy plus laser-ISL terminal energy; packets move through ISLs. It is not a payload-DC UE-handover result. Unlimited buffers prevent drops, not delay differences.
Fastenbauer et al., 2025	Fig. 15 reports almost 90% lower consumption at low load than full illumination.	Finite-demand scheduling and active-resource reduction can save modeled payload power. Eq. 28 includes PA efficiency, RF-chain/phase-shifter costs and a fixed floor. This is not a pure association ablation. At high load, differing treatment of infeasible users complicates energy-only comparisons.
Liao-Kaneko, 2021	Joint assignment/power optimization reports more than 99.7% lower "power consumption" against the maximum-power fixed-power comparator, with reduced sum rate.	The paper's figure terminology does not cleanly identify this headline as an RF-only reduction. Its model includes circuit power and constant PA efficiency. Fixed-power assignment is already optimized, so this does not isolate load-aware association or a third coordinator.
Cocco et al., 2018	Joint bandwidth/power allocation is evaluated through demand mismatch and fairness; additional power flexibility has limited benefit relative to bandwidth flexibility in the studied configurations.	Flexible payload resource matching is useful under shared TWTA constraints. The objective is not pooled EE; no EE percentage can be imported from those results.
Anzalchi et al., 2010; foundation	The reported comparison roughly halves TWTA DC consumption and reduces unmet demand with beam hopping.	Beam hopping can improve a modeled GEO architecture. Hardware sharing, bandwidth and carrier arrangements also change, so the result is not a clean isolated association or sleeping effect.

U — missing bridge. None of these papers measures the marginal of a third coordinator added to this project's two heads under the sealed capped-TDM law. No reviewed measurement calibrates the project's RF cap, PA curve, beam-to-chain mapping or fixed floors. The 2025 paper *Energy efficient LEO satellite communications: Traffic-aware payload switch-off techniques* was located (DOI 10.1016/j.comcom.2025.108122), but its primary manuscript could not be retrieved reliably; its quantitative claims are excluded.

2. Balancing versus consolidation

“Load” can mean simultaneous users, offered bit rate, beam demand, queue backlog, utilization or blocking risk. These lead to different optimization problems. Wang's balancing redistributes queued traffic across satellites while adapting beam hopping and power. Cocco's allocation balances demand satisfaction. Fastenbauer's scheduling exploits low offered traffic to reduce active-resource time. These are not interchangeable occupancy-variance objectives; the responding energy terms are listed above.

[Wang Cocco Fastenbauer](#)

The terrestrial foundation by Son et al. jointly addresses active base stations and user association with utilization-dependent power and delay. It reports a case with approximately 45% energy saving while delay remains below twice the all-on minimum-delay benchmark. This is direct evidence that energy savings can carry a quantified delay cost, but it is a terrestrial queueing model, not a LEO payload prediction. [Son et al., 2011](#)

Satellite beam hopping also has a temporal service cost even when aggregate demand is eventually met. The DVB-S2X implementation guide specifies switching guard/dummy symbols and receiver reacquisition support. They establish that illumination changes need waveform and timing treatment; they do not determine a universal interruption penalty. [DVB-S2X guide, Annex C.1.3](#)

A — the appropriate verdict is a tradeoff, not a slogan. For a baseline with delivered bits B_0 and positive energy E_0 , an alternative improves pooled EE exactly when $\Delta B - (B_0/E_0)\Delta E > 0$. Reducing both bits and joules can satisfy this inequality while degrading service. Conversely, expanding beams can increase bits enough to improve EE. Admission counts cannot establish rate attainment, delay, outage duration or continuity. Energy comparisons with unequal delivery remain interpretable as ratio comparisons only if those service differences are also disclosed.

In the old full-buffer, homogeneous singleton example, consolidation may reduce aggregate bits and energy together; therefore “fewer beams always improves EE” is false even before introducing target-rate power. In the successor, the relevant competition is increased required power and cap risk versus eliminated active-resource costs and changed interference. The user count alone does not resolve it. [A; P11/P20 for the respective model definitions]

U — application-level scope. The sealed successor continues to score achieved full-buffer ACM bits. Without a separately supplied arrival, queue and deadline model, it cannot establish useful application energy per delivered job or queueing-delay benefits. Literature with finite queues cannot silently supply that missing traffic model.

3. What a coordination component can add to per-agent values

3.1 Precedents and the strength of their ablations

Primary precedent	What the coordination mechanism and comparison establish	What remains unestablished
QMIX, Rashid et al., 2020	A monotonic mixer uses centralized information during training while preserving decentralized greedy action selection. §10 removes state conditioning and changes mixer form/capacity; contributions vary with task difficulty.	This is learned value factorization, not three prescribed physical reward heads. State access, representation and optimization effects cannot all be called an independent coordinator marginal. No satellite EE endpoint.
DCG, Böhmer et al., 2020	Pairwise payoff terms plus coordinated max-sum action selection beat a no-edge VDN baseline in tasks with costly miscoordination. Without the coordination penalty they can match VDN; on most studied SMAC maps gains are small or absent.	Explicit interactions can matter, but do not guarantee improvement. Communication, graph structure and omitted higher-order interactions limit transfer.
Weighted QMIX, 2020	Demonstrates that monotonic approximation can select the wrong joint argmax even when the underlying joint values are available; weighted projection changes this behavior.	Accurate labels or small regression error do not establish correct decisions. Tabular guarantees are not guarantees for deep networks or this handover model.
COMA, Foerster et al., 2018	A centralized critic supplies a counterfactual policy-gradient baseline. The same-actor comparison with a centralized QV critic isolates a credit-assignment contribution in tested combat tasks.	The baseline is a training device. It is not an extra deployed greedy Q-head, and does not justify independently executing all unilateral improvements.
Li et al., 2022	In mmWave multi-association, measured and enhanced difference rewards are compared in an EE/QoS utility. Exploration is sequential and measurement/control is coordinated.	Enhanced rewards include subsequent reactions, with information/protocol differences. This is not a clean third-head DROP under simultaneous LEO reassociation; higher EE can coexist with worse blockage performance.
Jang et al., 2026	Integrated LEO-terrestrial proactive handover compares QMIX-PHO with IQL-PHO in training reward; operational comparisons address throughput, handovers and ping-pong behavior.	Fig. 8 is a training-reward comparison, not an isolated EE endpoint. The reward addresses rate/continuity/shortfall; the paper supplies no payload-energy marginal.
DROO, Huang et al., 2020	A learned proposer generates complete offloading vectors; a model-based resource optimizer scores candidates and selects one.	This supports optimization-assisted proposal selection. The objective is computation rate, not satellite EE, and matching an optimizer with less computation is distinct from beating it in objective value.

3.2 Why accurate unilateral externalities can still fail

A. For a shared configuration objective F , a difference reward $D_u = F(a_u, a_{-u}) - F(c_u, a_{-u})$ preserves the ordering of user u 's alternatives when all other actions are fixed. It does not make the sum of separately selected differences equal the result when everyone changes. For two changes, the interaction is

$$\Psi = F_{11} - F_{10} - F_{01} + F_{00}.$$

Beam activation, final-user evacuation, interference and caps can produce nonzero Ψ . Allocating Ψ between users, even with an exact budget-balanced Shapley identity, does not guarantee both users adopt the complementary actions. This is a decision-composition problem as well as a credit-assignment problem. The distinction between aligned utilities and learnability has a foundational treatment in Wolpert and Tumer; DCG supplies a modern explicit interaction-and-decoder precedent. [Wolpert-Tumer DCG](#)

The package's MONE reversals, the learned LC-SRS gate passing while deployment fails, and the supplied probe's unilateral others-bit target fit these mechanisms. They are **P/U**, not external replications. Reading file 15 verifies what that included probe calculates; it does not verify the unavailable production tree or prove that historical R7 labels used the same calculation. R7 genuinely included two-user profiles, so the chronology cannot be summarized as eighteen purely unilateral experiments.

3.3 Best design prior and the meaning of “marginal”

A — design inference. The strongest prior is complete-configuration selection over physically coherent candidate outcomes, with atomic execution and explicit service constraints. Under file 20, a coordinator can exploit changes in required-power distribution, interference and activation that depend on multiple users. A learned contribution can improve proposal coverage, forecast future conditions or approximate an expensive evaluator. None of these roles requires naming occupancy variance as an independent objective.

C1 already owns unilateral whole-network surplus in the successor. Assigning those same energy changes again to C3 double-counts them. C3's distinct claim therefore concerns remaining interactions, anticipation or search capability. C2's future contribution must likewise be distinguished from current-slot utility; adding values with different horizons or scales does not acquire a common objective merely because they are Q-heads. [A; P20]

The reviewed ablations imply four different estimands: removing a module from one checkpoint measures dependence of that trained policy; retraining without it measures the architecture's learning contribution; comparing joint and unilateral candidate selection measures interaction/search capability; comparing learned and equally informed model-based selection measures added learning value. These are analytical distinctions, not a new experimental protocol. No one DROP answers all four. A strong baseline may legitimately absorb the gain; DCG's near-ties and QMIX's easier-task ablations make that possibility empirically credible. [DCG QMIX](#)

If the baseline configuration is included in a candidate set, maximizing the same nominal immediate objective cannot score below that baseline in that evaluator. **This is only an A-level, catalogue-scoped statement.** It does not guarantee realized fading performance, continuing-system EE, service preservation under model error, or improvement over an unrestricted optimizer. “Oracle” must specify its information, catalogue, horizon, constraints and decoder.

4. Reward-objective alignment for EE ratios

4.1 A ratio endpoint is not a sum of ratios

A. For positive interval energies,

$$\eta_{\text{pooled}} = (\sum_t B_t) / (\sum_t E_t) = \sum_t [E_t / (\sum_j E_j)] \times (B_t / E_t).$$

Pooled EE is an energy-weighted mean of interval EEs. An unweighted sum or mean of per-step ratios gives different weights, which policies themselves can change. Similarly, averaging per-user or per-world ratios is generally a different objective from aggregating their bits and joules. Correct instantaneous summation over users does not fix incorrect aggregation over time. Discounting introduces another endpoint distinction. These are identities, not empirical claims about a particular implementation.

For any fixed comparison baseline, $\eta_0 = B_0/E_0$ gives

$$\eta_1 > \eta_0 \text{ if and only if } \Delta B - \eta_0 \Delta E > 0.$$

A — correction to an overstrong interpretation in the package. η_0 need not be the globally optimal ratio for this exact baseline-comparison identity. An arbitrary frozen price, however, does not preserve every pairwise EE ordering or turn local action surpluses into a guarantee on a closed-loop pooled endpoint.

Dinkelbach's foundational fractional-programming method uses successive subtractive problems and updates the ratio parameter. Shen and Yu explain its communications application and distinguish single-ratio transformations from more general coupled problems. Calling a fixed training reward $B - \lambda E$ “Dinkelbach” does not establish that the iteration, inner optimization or optimality conditions have been satisfied. Nonconvex interference and approximate RL introduce additional limits. [Dinkelbach, 1967](#)
[Shen-Yu, 2018](#)

4.2 Difference rewards do not repair the wrong global objective

A. A perfectly estimated difference of an unweighted step-ratio objective remains aligned to that objective, not automatically to pooled EE. A counterfactual baseline can improve credit assignment without changing the actor's intended policy-gradient objective, as COMA demonstrates. It cannot supply a missing energy term, unavailable same-slot information or coordinated action adoption. [COMA](#)

Jin et al.'s fractional deep RL paper provides a networking example of distinguishing a ratio of expectations from the expectation of a ratio for age of information. Its NonFrac versus fractional comparison is relevant to objective alignment; a larger combined-method headline also changes the action formulation and cannot be attributed only to fractional rewards. It is an AoI/mobile-edge result, not an EE coordination-head failure. [Jin et al., 2024](#)

U — requested documented case. No reviewed primary study was verified to show that a third satellite EE coordination component looked useless specifically because per-step ratio rewards disagreed with a pooled EE endpoint. The algebra establishes that this can occur in principle; the closest verified networking precedent concerns fractional AoI. It would be inaccurate to present the project's diagnosis as an already established published satellite case.

For this package, the stale live-branch λ , ownership of nonfocal bits, the legacy bootstrap and checkpoint objective are separate issues. File 18 reports correct stored-label prices and a correctly summed instantaneous r_1 , so those findings narrow earlier accusations. Its per-head bootstrap PASS addresses head/target-network routing; file 12's objection concerns taking separate next-action maxima instead of

one deployable scalarized action. Those are different tests. The latter is not resolved by the former. Production behavior remains P/U without the full producing tree.

5. Simulation validity: what is documented, and what is not

Sargent's foundational verification-and-validation framework separates conceptual-model validity, implementation verification, data validity and operational validity. It describes analytical comparisons, extreme/degenerate conditions, event tracing and independent review. This directly refutes treating arithmetic reconciliation alone as physical validation. It does not establish that wireless-RL papers routinely apply every such technique. [Sargent, 2010](#)

There are concrete wireless calibration precedents. Sandri et al. implement an NTN channel model in ns-3 and calibrate against 3GPP reference behavior. Koutlia et al. calibrate 5G-LENA against 3GPP reference scenarios using intermediate coupling-gain/SINR distributions and system outputs, reporting remaining discrepancies. These support reference-based link/channel calibration; neither validates a LEO PA-energy model or this handover procedure. [Sandri et al., 2023](#) [Koutlia et al., 2022](#)

Practice asked about	Primary-source status	What it can establish
Independent implementation / model reconstruction	Independent V&V and analytical/model comparison are documented in Sargent ; the package reports its own replay in file 18.	Agreement checks implementation of a specification. Shared conceptual errors can survive; full independent wireless-RL reimplementaion is not established as routine by this review.
Known-answer and limiting cases	Analytic, degenerate and extreme-condition checks are documented by Sargent .	They discriminate implementation faults for the exercised cases, not all regimes or the physical truth of a formula.
Differential link-budget calibration	Concrete wireless/NTN precedents in Sandri and Koutlia .	Comparison of intermediate physical quantities localizes discrepancies better than endpoint-only agreement. Calibration domains still limit generalization.
Placebo / null arms	Same-algorithm split comparisons appear in Colas et al. ; RL evaluation protocol artifacts appear in Agarwal et al. .	A null is useful for measuring false discoveries. No verified primary source here establishes full coordinator-path placebo tests as standard wireless practice.
Invariance and mutation tests	General special-condition reasoning is supported by Sargent ; the exact project invariances are A-level consequences of its declared model.	A same-state history-independent controller should not change with irrelevant association history. This is a logical contract, not a sourced hardware measurement. Mutation-test prevalence in wireless RL is U.

Two standards resolve particular register claims. ITU-R S.465-6 provides a reference earth-station pattern only over its specified angular domain; it does not validate the project's below-domain extrapolation or plateau. 3GPP TS 38.331 Release 18, §5.5.4.15a, Event D2 combines serving-distance and candidate-distance conditions. A candidate-only gate is therefore D2-derived, not the complete standardized event. Neither standard quantifies the effect of the project's deviations on EE. [ITU-R S.465-6](#) [3GPP TS 38.331](#)

P/U — what file 18 changes. It reports an independently written replay over its declared panel, with close agreement except a small Bessel-related bit discrepancy and an unused S.465 branch issue. That reduces the evidence for a gross arithmetic error in those exercised formulas. The replay deliberately retains private wanted power versus beam-max interferer power: agreement therefore does not resolve that transmission-model inconsistency. Nor does a bounded replay discrepancy certify a bound on policy differences near action thresholds. The historical SINR summaries used different panels; later interference diagnostics do not by themselves establish an exact paired causal explanation for the old discrepancy.

U — published gain artifacts. This review did not verify a primary-source case attributing a published satellite EE or handover gain specifically to an erroneous power or handover model. Agarwal et al. document apparent RL gains sensitive to evaluation protocol; that is a relevant warning, not the requested physical-model case. The absence of a verified case here is not evidence that such artifacts never occur. The project's re-association premium remains a supplied model diagnosis with unmeasured attribution, not a replicated publication correction.

6. Detecting or excluding a 1-3% pooled-ratio marginal

There is no documented universally sufficient seed count. Colas et al. tie detectability to effect size, variance, error rates and the reliability of variance estimates; small pilots can mislead. Agarwal et al. show uncertainty and ranking instability in few-run RL evaluations, including bootstrap coverage limitations. Neither study certifies a design for this particular pooled ratio or justifies treating many correlated simulator steps as independent replications. [Colas et al.](#) [Agarwal et al.](#)

The relevant documented design principles are paired comparisons, preservation of randomization/dependence units, ratio-aware covariance calculations and explicit meaningful-effect bounds. Common random numbers can reduce comparison variance under appropriate coupling conditions, but their benefit is not automatic. World pairing controls common exogenous conditions; different endogenous trajectories are part of the policy effect, not a reason to force identical visited states. [Glasserman-Yao, 1992](#); [foundational, abstract-level access](#)

6.1 The estimand and its uncertainty

A — ratio-aware derivation using the delta method. Let an independent paired outer block contain total bits and joules (B_F, E_F, B_D, E_D) for FULL and DROP. With positive population means, define

$$L = \log(\mu_{BF}) - \log(\mu_{EF}) - \log(\mu_{BD}) + \log(\mu_{ED})$$

$$\psi = (B_F/\mu_{BF} - E_F/\mu_{EF}) - (B_D/\mu_{BD} - E_D/\mu_{ED}).$$

For n IID paired outer blocks with adequate finite moments, the asymptotic variance of the estimated L is approximately $\text{Var}(\psi)/n$. This retains within-arm bit/energy covariance and between-arm pairing. It estimates the log contrast of pooled ratios, not the average episode log-ratio. Individual blocks may have zero bits; the aggregate/population means used in the logarithms must be positive. This exact paired-log expression is a derivation here; the general ratio/covariance method is supported by Deng et al. [Deng et al., 2018](#)

If worlds, orbital-date blocks and training lineages are reused across cells, the cells are not IID outer blocks. Crossed resampling or covariance treatment must follow the actual dependence structure; simply calling all cells “seeds” is inadequate. Owen and Eckles establish multiway bootstrap results for crossed arrays, principally means under stated assumptions. Extending that logic to this smooth ratio requires additional asymptotic reasoning; it is not an exact small-sample guarantee. More world trajectories with fixed trained policies do not establish robustness to new training runs. [Owen-Eckles, 2012](#); [foundation](#)

6.2 What different claims require

Intended statement	Necessary interpretation of a valid interval for relative pooled EE
Detect a positive marginal	Its lower bound is above zero.
Exclude a gain at least δ	Its upper bound is below the prespecified meaningful gain δ . The user's 1–3% range contains distinct possible thresholds; the claim must identify which one.
Establish equivalence	The appropriately constructed interval lies inside prespecified lower and upper equivalence bounds.
Report a finite-panel result	Exact pooled arithmetic describes that panel. It does not alone support a population or training-distribution claim.

These distinctions follow the logic of equivalence testing; failing to reject zero does not demonstrate negligible effects. A small positive effect can be detected while a larger meaningful effect is excluded. The interval's coverage and the threshold must be justified before interpreting either claim. [Lakens, 2017](#)

U — sufficiency for this project. The package does not provide the independent-block variance and dependence evidence needed to establish precision for the requested effect range. No literature result can replace those quantities. Few training lineages remain a limitation even with many worlds; few independent clusters also limit bootstrap reliability. Repeatedly inspecting panels, choosing thresholds afterward or selecting the best checkpoint/contrast adds selection uncertainty. This review therefore cannot certify that a negative old screen excludes a 1–3% successor gain, and it proposes no new seed counts or parameter settings.

7. Verdicts on the project's root-cause hypotheses

7.1 File 16 §B: all ten ranked hypotheses

The order below follows file 16's ranking; parenthetical IDs preserve its original hypothesis numbering. The subjective probabilities in that file are not literature-derived and are not adopted. All project observations in this table are P/U.

Rank / hypothesis	Literature verdict	Project-specific conclusion and remaining limit
1 (3): unilateral/additive mismatch	Supported mechanism. Difference utilities and joint-action decoding are distinct; non-additivity can defeat local greedy composition. DCG , Weighted QMIX .	MONE reversals fit this explanation. R7's positive teacher-composition result prevents a claim that every additive policy is incapable. Contribution to each failure is unverified.
2 (1): missing occupancy → required-power	Supported narrowly by A. Rate targets introduce the coupling; max-RF full-buffer sharing does not generally do so.	"No coordination lever at all" is contradicted by the supplied activation/interference model and positive search results. Successor headroom is unmeasured.
3 (9): information / learnability bottleneck	Supported possibility. Central state and counterfactual information matter; representation quality is distinct from decision quality. QMIX , COMA .	Hidden-fading ZR cannot establish deployable value. Nominal S0 and the R7 learner pass prevent a universal information-impossibility claim.
4 (4): executed oracle differs from declared C3	Validity objection supported. A solver must correspond to its specified model/objective. Sargent .	Included file 15 calculates unilateral nonfocal bit changes, not the declared interaction Ψ . That direct code reading concerns this probe only, not all historical coalition labels.
5 (2): balancing opposed consolidation incentives	Conditional support. Activation savings and target/queue pressure can pull in opposite directions. Fastenbauer , Son .	ZR's reported lower bits and energy fit consolidation. Neither R7 nor the literature supports "always consolidate."
6 (5): training / target-stack defects	Supported class of failure. Ratio transformations require matching objectives; accurate projected values need not select the right action. Shen-Yu , Weighted QMIX .	File 18 narrows label-price/r1 accusations but does not resolve the separate bootstrap action issue. Supervised heads bypass some legacy TD defects. No single defect explains every attempt.
7 (8): physics implementation errors	Supported concern, unproved attribution. Calibration and code verification have limited domains. Sandri , Sargent .	File 18 lowers gross arithmetic-error concern for its panel; faithful replay leaves the radiation-model inconsistency intact. No measured causal contribution to C3 failure.
8 (10): restricted proposals, horizon, service	Supported limitation. Coordinated solvers and proposers only search their represented actions; energy-delay tradeoffs require service semantics. DROO , Son .	Finite-catalogue negatives cannot exclude omitted joint moves or longer-term value. Their existence or benefit in this system is U.
9 (6): small effects statistically missed	Supported possibility, not an explanation of all signs. Few-run uncertainty is real. Colas , Agarwal .	Deterministic comparisons already record finite-panel signs. Uncertainty limits generalization; it does not turn negative arithmetic into positive evidence.

Rank / hypothesis	Literature verdict	Project-specific conclusion and remaining limit
10 (7): BASE absorbs all gains	Supported possibility. Coordination gains depend on the task and comparator. DCG .	Historical S0 and unilateral search positives contradict an exhausted learned BASE on those panels. A retrained successor could absorb them; that outcome is U.

7.2 File 11 §1: complete H01-H43 coverage

“Supported concern” means the concern follows a cited source or the stated algebra. It never means that the reported code behavior or its effect size has been externally reproduced. Rows marked U have no verified primary source for the specific claim or causal magnitude.

ID / assumption challenged	Verdict and exact support	Unverified boundary
H01 — entry history controls RF/feasibility	A: memoryless state control must not inherit an arbitrary entry anchor. Only the specified transmit-gain product is held by inversion, not all received power or throughput.	P/U: entry-history implementation and any fraction of the reported 2.9% gain. Power need not rise monotonically.
H02 — private wanted power with beam-max radiation	Supported consistency concern: the scheduled transmitter must have coherent desired/interference/energy accounting; compare explicit scheduling models in Fastenbauer .	P/U: no reviewed primary source validates the project's hybrid transmitter. Effect magnitude unknown.
H03 — failed links disappear before radiation	A: a scheduled failed attempt still has its transmission energy and interference; file 20 explicitly retains it.	P/U: exact old pruning path and its causal importance. No external source authenticates that code.
H04 — a snapshot represents a long interval	Supported concern: dynamic NTN geometry/channel modeling has explicit time semantics. Sandri .	U: adequacy or bias of the project's chosen interval is not established by a primary source.
H05 — cached identities equal minimum residence	A: candidate caching and a prohibition on switching are different constraints.	P/U: the reported stale clock and cache behavior; no literature-based magnitude.
H06 — short priming reconstructs latch history	A: a hysteretic state requires sufficient causal history to identify its current state.	P/U: whether the implemented priming reaches the correct state for each start.
H07 — elevation configuration is live	Verification concern: advertised configuration must reach the model. Sargent .	P/U: reported dead configuration and resulting eligibility changes.
H08 — initialization is horizon/order independent	A: scored horizon should not alter an otherwise identical declared initial physical state; RNG dependence requires explicit treatment.	P/U: reported prefix/order effects. A seed label alone proves neither independence nor causality.
H09 — observed candidate SINR equals execution CSI	Supported concern: state conditioning changes learnability. QMIX . A: independent observation noise does not reveal realized execution fading.	P/U: mixed timestamps and exact information gap; no primary source measures this project.

ID / assumption challenged	Verdict and exact support	Unverified boundary
H10 — Shannon rate is realizable ACM	Refuted as an identity: standardized finite modes/reference thresholds differ from Shannon capacity. DVB-S2 .	U: actual modem profile and implementation loss are not calibrated here.
H11 — Q1 + Q2 equals system surplus	A: omitted nonfocal bits break the claimed identity; difference rewards do not automatically sum to F. Wolpert-Tumer .	P/U: complete historical producer semantics and scale compatibility.
H12 — old λ and κ transfer unchanged	Supported concern: the ratio price relates to a specified objective/reference. Shen-Yu . A: an outage preference is not merely a units conversion.	U: correct successor calibration; none is proposed.
H13 — bootstrap/checkpoint objective matches deployment	A: separate head maxima can represent incompatible next actions.	P/U: file 18's network-routing PASS does not settle file 12's action-selection objection; full production code unavailable.
H14 — empty masks/outages can be censored	A: deleting elapsed time or failures changes the evaluated process and endpoint.	P/U: where censorship occurs and its magnitude.
H15 — ideal sleep and omitted floors are harmless	Supported concern: active-resource/fixed-power terms determine sleeping savings. Fastenbauer . A: a common floor can change ratio rankings.	U: actual hardware switchability, wake costs and floors.
H16 — Φ is physical handover energy/time	A: a preference penalty has no joule/second interpretation without a physical model.	P/U: repriced heads reportedly omit Φ ; attributing their result to Φ -training is contradicted by the package correction.
H17 — served fraction protects QoS	Refuted as sufficient: energy and delay are distinct outcomes. Son ; ACM can preserve connectivity with reduced rate. ITU-R S.2131 .	U: application QoS acceptability of the project's guard.
H18 — four fields describe deployed C2	Verification concern: feature consumers must match documented semantics. Sargent .	P/U: the reported Q1/Q2 feature routing; external literature cannot authenticate dimensions or code paths.
H19 — additive, set and learned coordination are equivalent	Refuted: interactions and coordinated decoding differ from additive values; learned proposal scoring also differs. DCG , DROO .	P/U: how much historical gain came from unilateral versus genuinely joint moves.
H20 — reused development worlds are fresh validation	Supported concern: evaluation and selection protocols can alter apparent RL gains. Agarwal .	P/U: exact history of opened worlds; future generalization remains unknown.
H21 — DROP is an independent main effect	A: removing one component measures a conditional contrast; interaction and retraining define different estimands.	U: independent causal contributions without the corresponding design and lineage coverage.
H22 — an exact positive inequality proves statistical support	Refuted: finite-run uncertainty needs explicit inference. Colas , Deng .	U: precision for the desired pooled effect range.

ID / assumption challenged	Verdict and exact support	Unverified boundary
H23 — short resets establish continuing value	A: terminal conditions and initial-state distributions define the return being estimated.	U: size and direction of the project's horizon bias; not established by a primary source.
H24 — receipts/verifiers prove physical correctness	Refuted: verification and operational validity are distinct. Sargent .	P/U: source authentication and shared-model errors remain separate questions.
H25 — search latency is accurately measured and free	Supported concern: DROO explicitly treats solver cost as part of practical method value. DROO .	P/U: corrected historical timing is reported; deadline-tail reliability and physical service cost are unverified.
H26 — inherited RF cap is derived hardware	U: no reviewed primary source derives this cap for the modeled payload.	Compatibility with a total budget does not identify a hardware operating limit.
H27 — square-root PA efficiency validates a payload	Refuted as validation: amplifier operating point and distortion need explicit models. DVB guide , Cocco .	U: the project's PA supply curve and back-off calibration.
H28 — circuit coefficients identify beam hardware	Supported concern: shared amplifiers and active-chain mappings matter. Cocco , Fastenbauer .	U: one-beam/one-chain mapping and the project's component coefficients.
H29 — PA/baseband must be charged per user	A: physical devices are charged once per actual operating schedule/inventory, not multiplied by user count.	P/U: once-per-beam/satellite is coherent only with the declared physical mapping; its hardware truth is U.
H30 — full-buffer capacity equals useful application delivery	Refuted as general identity: finite-demand studies explicitly distinguish offered and served traffic. Fastenbauer .	U: application arrivals, usefulness and deadlines in this project.
H31 — equal sharing is FDM / consolidation must win	A: full-band TDM keeps full-band noise during each transmission; both bits and energy may fall on consolidation.	U: no primary source supplies universal consolidation dominance.
H32 — reuse pattern / closed interferers are universal	Supported concern: interference topology is a system-model assumption. Wang .	U: external traffic and real reuse/beam alignment for the proposed constellation.
H33 — S.465 defines the near-axis extrapolation	Refuted: its reference pattern has an explicit angular applicability domain. ITU-R S.465-6 .	U: below-domain antenna behavior and EE effect of replacing the extrapolation.
H34 — ideal transmit nulls / numerical routing are hardware truth	Supported verification/validity distinction. Sargent .	U: measured beam pattern. P/U: replay's numerical agreement does not validate ideal nulls or global policy robustness.
H35 — clear-sky / median shadowing is all-weather expectation	Refuted as identity: throughput statistics and ACM require the relevant distribution. ITU-R S.2131 .	U: omitted weather/blockage distributions and their policy effect.
H36 — common random numbers make fading realistic	Refuted: CRN is a comparison-variance technique, not a channel-correlation model. Glasserman-Yao ; A for the distinction.	U: physical spatial/temporal correlation and observability.

ID / assumption challenged	Verdict and exact support	Unverified boundary
H37 — TDM thermal noise needs occupancy division	A: full-band scheduled reception uses NoW; dividing noise alone changes the link model.	No primary source is needed for this dimensional/model identity; actual receiver noise calibration remains U.
H38 — nearest TLE is necessarily causally available	A: nearest-in-time may be future information; retrospective and operational information sets differ.	U: project ephemeris availability and the resulting benefit. No orbit-accuracy claim is made.
H39 — Earth/frame approximations are negligible	U: no verified primary source bounds their effect for this geometry and decision thresholds.	General channel calibration Sandri does not establish this specific bound.
H40 — coarse shortlist is a safe superset	A: a restricted candidate catalogue cannot certify omitted actions.	U: completeness of the project's shortlist; no literature result proves it.
H41 — candidate-only D2 is the full standardized event	Refuted: Release-18 D2 includes serving and candidate distance conditions. 3GPP TS 38.331 .	P/U: actual gate behavior; no quantified EE consequence from the standard.
H42 — mobility/Doppler need no receiver assumptions	Supported modeling concern: NTN channel/mobility assumptions must be explicit. Sandri .	U: real tracking impairment; raw Doppler alone does not establish decoder loss under assumed ideal compensation.
H43 — geometry masks prove hardware capacity	Refuted as sufficiency: flexible payloads have coupled physical resource constraints. Cocco .	U: beam inventory, feeder/processing bottlenecks and operational feasibility of every allowed assignment.

7.3 Chronology: all eight families and eighteen indexed rows

The chronology calls itself a derived index and spans 31 August–7 September for these entries. It is not eighteen independent replications or the entire three-week work history. The table preserves its family mapping; figures are supplied historical reports, not recomputed evidence.

Family / rows	Hypothesis tested and reported outcome	Literature verdict and limit
A / 1 — masked mean/max	Matched-opening nonfocal externality; validation/estimand problems; recorded unresolved, not refuted.	Credit learnability and deployment effect differ. Wolpert-Tumer . A failed gate does not establish no physical mechanism. P/U.
B / 2 — PNFE	"Others hold" committed background; reported EE –11.06%.	A: an externality conditional on fixed others need not predict simultaneous changes. Better estimation of that conditional target cannot repair the counterfactual mismatch. P/U attribution.
C / 3–4 — MONE and joint-response sweep	MONE: bits +5.373%, energy +8.227%, EE –2.637%; positive summed predictions in all 60 diagnostics, 28 joint reversals.	Non-additivity is consistent with DCG and A-level Ψ . An ordered sweep is not a general joint optimum. Activation dominance is a package interpretation, not externally measured causation.

Family / rows	Hypothesis tested and reported outcome	Literature verdict and limit
D / 5-7 — ZR oracle/learner	Fresh ZR approximately +0.94%, with both bits and energy lower; learner failed.	Energy-support gating can select consolidation without identifying the named externality. Hidden simultaneous support and estimation loss can differ from achievable decision value. QMIX ; P/U for the particulars.
E / 8-11 — context, origin, loss	Re-centering/context rescue rejected; added origin information and loss changes still failed.	A: action-independent shifts preserve argmax; unchanged information cannot reveal independent hidden variables. The package's distributional explanation is not proof of a universal representation limit. P/U.
F / 12-13 — relational/normalized	Cache digest mismatch produced an instrument STOP; learner gates also failed.	Sargent supports separating verification from model efficacy. Floating-point nonidentity is not automatically physically meaningful disagreement. Subsequent learner failures remain distinct P/U outcomes.
G / 14-15 — repriced/expected ZR	Exact-fading arm remained about +0.985%; nominal weaker; expected-fading screen lost support.	Fixed-price mismatch alone is not a sufficient explanation of this history. Oracle information matters; exact realized fading is not deployable CSI. Shen-Yu , A; P/U causal limit.
H / 16-18 — LC-SRS and D/F	R7 physical pair -0.049%; teacher composition +0.335%; learned composition -0.660% despite learner-gate PASS. D/F fast screens also negative.	Budget balance, physical headroom, decoding and learning are separate. Weighted QMIX , DCG . Restricted pair/catalogue negatives do not exhaust coordination; a two-step screen does not exclude a population 1-3% effect. P/U.

Later package evidence changes the interpretation without resolving causation. S0 reports +1.813%; its unilateral and joint variants are not nested interventions from which a clean synergy can be obtained by subtraction. E1 reports unilateral +1.992% and joint +2.222% over its finite realized-information catalogue. C3-S FULL/LITE report approximately +2.883%/+2.922% on a TRAIN panel. These outcomes establish reported restricted opportunities against the respective baselines, not deployable successor gains or pure multi-user learning value. [P04/P05/P06; P/U]

O1's reported +8.465% also has a later live-branch price caveat. File 11 retracts earlier claims that the repriced heads were trained with Φ , that phase equals association age, or that anchoring freezes throughput. The approximately 2.9% arithmetic can be sound while its mechanism remains unidentified. A selective history-reset incentive is plausible under the old recurrence; its sign and causal share are not established. [P03/P06/P11/P18; P/U and A]

8. Annotated primary references

Each entry states the exact claim used in this review and its ceiling. Section/figure/equation locators refer to the linked primary text. The annotations do not convert model-based simulations into payload measurements.

R1 — ETSI (2014), EN 302 307-1 V1.4.1, DVB-S2. Standard, explicitly retained before 2015. Official standard. §6/Table 13 supports finite MODCOD efficiencies and reference decoding thresholds; §5.6 specifies roll-off. Refutes Shannon/ACM identity. Does not supply the project's implemented modem or universal real-channel decoding performance.

R2 — ETSI (2015), TR 102 376-1 V1.2.1, DVB-S2 implementation guidelines. Official report. §4.1.2 permits power control/ACM; §§4.3.2.1–2 discuss amplifier and carrier effects. Supports treating RF control and PA operating assumptions explicitly. Does not validate a generic square-root PA curve or per-user LEO control implementation.

R3 — ITU-R (2019), Recommendation S.2131-0, performance objectives for satellite links using ACM. Standard. Official recommendation. Recommendations and note 4 support availability/throughput-distribution treatment. Refutes equating maintained connectivity or a nominal link point with constant achieved throughput. No project hardware or traffic calibration.

R4 — ITU-R (2010), Recommendation S.465-6, reference earth-station radiation pattern. Standard, explicitly retained before 2015. Official recommendation. Recommends 2 and the minimum-angle conditions support the domain objection to near-axis extrapolation. They do not prescribe the project's below-domain plateau or quantify an EE effect.

R5 — 3GPP/ETSI (2024), TS 38.331 Release 18, V18.1.0, NR RRC protocol specification. Standard. Official specification. §5.5.4.15a defines Event D2 using serving and candidate distance conditions. Refutes calling a candidate-only selector the full event. No direct energy model or handover-cost calibration.

R6 — ETSI (2015), TR 102 376-2 V1.1.1, DVB-S2X implementation guidelines. Official report. Annex C.1.3 specifies beam-hopping switching alignment, guard/dummy symbols and reacquisition support. Supports nonzero waveform/timing considerations. Does not give a universal service penalty for this simulator.

R7 — Wang, G.; Yang, F.; Liu, J.; Song, J.; Han, Z. (2026), “Joint Resource Management and Load Balancing in Multi-Satellite Beam Hopping With Interference Suppression: An Energy Minimization Perspective.” IEEE TWC 25, 8131–8147. DOI 10.1109/TWC.2025.3635684. Original paper text. Eqs. 9/11 and §V/Figs. 6–9 establish the RF-plus-ISL energy boundary and no-balancing comparison. Supports one load-balancing marginal; not a UE-association payload-DC or learned-head marginal.

R8 — Fastenbauer, A.; El Hadfi, Y.; Kaneko, M.; Svoboda, P.; Rupp, M. (2025), “LEO Satellite Beam Hopping for Power Consumption Minimization at Different Elevation Angles.” IEEE OJCOMS 6, 6930–6952. DOI 10.1109/OJCOMS.2025.3602090. Original paper text. §II.I/Eq. 28 and Fig. 15 support modeled active-resource energy savings; high-load feasibility handling limits comparisons. Does not isolate association or validate flight hardware.

R9 — Cocco, G.; De Cola, T.; Angelone, M.; Katona, Z.; Erl, S. (2018; online 2017), “Radio Resource Management Optimization of Flexible Satellite Payloads for DVB-S2 Systems.” IEEE TBC 64(2), 266–280. DOI 10.1109/TBC.2017.2755263. Author manuscript. §II and §V.B support shared TWTA, operating-point and demand-matching constraints. Objective and conclusions concern unmet/excess capacity and fairness, not an isolated EE gain.

R10 — Anzalchi et al. (2010), “Beam Hopping in Multi-Beam Broadband Satellite Systems: System Simulation and Performance Comparison With Non-Hopped Systems.” ASMS/SPSC. DOI 10.1109/ASMS-SPSC.2010.5586860. Foundational satellite comparison. Author manuscript. Table 1 supports its reported TWTA-DC/unmet-demand improvement. Multiple architecture changes prevent assigning the effect solely to user association or sleeping.

- R11 — Liao, Q.; Kaneko, M. (2021), “Global Energy Efficiency Optimization of a Ka-Band Multi-Beam LEO Satellite Communication System.” IEEE Access 9, 55232–55243. DOI 10.1109/ACCESS.2021.3071475. [Author manuscript](#).** Eqs. 7–10 and §V.C support the circuit-plus-constant-efficiency power model and optimized-power versus fixed-power comparison. At most one scheduled user per beam; not the successor's multi-user capped-TDM model.
- R12 — Son, K.; Kim, H.; Yi, Y.; Krishnamachari, B. (2011), “Base Station Operation and User Association Mechanisms for Energy-Delay Tradeoffs in Green Cellular Networks.” IEEE JSAC 29(8), 1525–1536. Foundational terrestrial analogy. [Author manuscript](#).** §II.C/Eq. 9 and numerical results connect association, utilization-dependent power, active BSs and delay. Supports a quantified energy-delay tradeoff, not a LEO savings estimate.
- R13 — Rashid et al. (2020), “Monotonic Value Function Factorisation for Deep Multi-Agent Reinforcement Learning.” JMLR 21(178), 1–51. [Primary paper](#).** §10/Figs. 12–13/17 separate aspects of mixer state and architecture. Supports task-dependent benefits of centralized value mixing. Does not establish an independently additive physical coordination reward or EE gain.
- R14 — Böhmer, W.; Kurin, V.; Whiteson, S. (2020), “Deep Coordination Graphs.” ICML, PMLR 119. [Primary paper](#).** §5.1 and §5.4 compare explicit pairwise coordination with VDN and show both gains and near-ties. Supports interaction-aware decoding when miscoordination matters; not universal improvement or a complete higher-order solver.
- R15 — Rashid, T.; Farquhar, G.; Peng, B.; Whiteson, S. (2020), “Weighted QMIX: Expanding Monotonic Value Function Factorisation for Deep Multi-Agent Reinforcement Learning.” NeurIPS. [Primary paper](#).** Projection analysis and examples show that value approximation can select a wrong joint action. Supports the non-additivity/decision-alignment objection, not a guarantee for deep satellite policies.
- R16 — Foerster, J.; Farquhar, G.; Afouras, T.; Nardelli, N.; Whiteson, S. (2018), “Counterfactual Multi-Agent Policy Gradients.” AAAI. [Primary paper](#).** §4/Eq. 4 and §5 establish a counterfactual actor-gradient baseline and same-actor critic comparison. Supports credit-assignment benefit in tested tasks; no claim of additive simultaneous greedy control or payload EE.
- R17 — Li et al. (2022), “Multi-Agent Reinforcement Learning With Measured Difference Reward for Multi-Association in Ultra-Dense mmWave Network.” IEEE Access 10, 118747–118758. DOI 10.1109/ACCESS.2022.3221455. [Original paper text](#), [ResearchGate copy](#).** §§III.C–IV support measured difference rewards and sequential exploration for an EE/QoS utility. Protocol and response-model differences limit a pure coordinator attribution.
- R18 — Jang et al. (2026), “Proactive handover optimization via multi-agent deep reinforcement learning in integrated LEO satellite-terrestrial networks for high-mobility terminals.” ETRI Journal 48, 577–589. DOI 10.4218/etrij.2025-0240. [Primary article](#).** §3.2 and Fig. 8 support the reward definition and QMIX/IQL learning comparison. Operational handover results concern throughput/continuity, not energy.
- R19 — Huang, L.; Bi, S.; Zhang, Y.-J. A. (2020; online 2019), “Deep Reinforcement Learning for Online Computation Offloading in Wireless Powered Mobile-Edge Computing Networks.” IEEE TMC 19(11), 2581–2593. [Author manuscript](#).** §IV.B/Eq. 10 supports neural candidate generation followed by resource optimization and selection. Supports the hybrid architecture prior and computational value; no satellite EE claim.
- R20 — Shen, K.; Yu, W. (2018), “Fractional Programming for Communication Systems—Part I: Power Control and Beamforming.” IEEE TSP 66(10), 2616–2630. [Author manuscript](#).** §II.A.2/Eqs. 5–6 explain Dinkelbach's single-ratio iteration; later coupled applications clarify optimization limits. Supports the distinction between a fixed surrogate price and a fractional optimization method.
- R21 — Dinkelbach, W. (1967), “On Nonlinear Fractional Programming.” Management Science 13(7), 492–498. DOI 10.1287/mnsc.13.7.492. Foundational work. [Publisher record](#).** Names the original nonlinear fractional-programming method. Access here was to primary metadata/abstract; detailed method claims are verified through Shen–Yu, not an asserted full reading of the 1967 text.

R22 — Jin, L.; Tang, M.; Zhang, M.; Wang, H. (2024), “Fractional Deep Reinforcement Learning for Age-Minimal Mobile Edge Computing.” *AAAI* 38(11), 12947–12955. DOI 10.1609/aaai.v38i11.29192. [Author manuscript](#). Objective formulation and NonFrac/Frac comparisons support networking relevance of ratio-of-expectations alignment. Its endpoint is Aol; the headline OFL+U comparison also adds updating/waiting control, so its gain cannot be attributed solely to fractional reward handling.

R23 — Wolpert, D. H.; Tumer, K. (2002), “Collective Intelligence, Data Routing and Braess' Paradox.” *JAIR* 16, 359–387. **Foundational work.** [Primary article](#). Develops aligned private utilities and learnability as distinct concerns. Supports difference-utility reasoning; not a theorem that local greedy actions compose optimally. Later manuscript upload dates are not the publication year.

R24 — Sargent, R. G. (2010), “Verification and Validation of Simulation Models.” *Winter Simulation Conference*, 166–183. **Foundational methodology.** [Primary paper](#). §§1–4 distinguish forms of validity and describe analytical, extreme-condition and independent checks. Supports limits of receipts/replay; not evidence that every wireless-RL paper uses these practices.

R25 — Sandri, M.; Pagin, M.; Giordani, M.; Zorzi, M. (2023), “Implementation of a Channel Model for Non-Terrestrial Networks in ns-3.” *WNS3*. DOI 10.1145/3592149.3592158. [Author manuscript](#). Model sections and §4 calibration support a concrete NTN reference-comparison precedent. No payload DC, project orbit-error bound or complete handover validation.

R26 — Koutlia, K.; Bojovic, B.; Ali, Z.; Lagén, S. (2022), “Calibration of the 5G-LENA System Level Simulator in 3GPP reference scenarios.” [Primary manuscript](#). §§3.4–5 compare coupling gain, SINR and system outputs with reference calibration behavior. Supports intermediate-quantity validation and residual-domain limits; terrestrial full-buffer results do not validate LEO energy or handover.

R27 — Colas, C.; Sigaud, O.; Oudeyer, P.-Y. (2018), “How Many Random Seeds? Statistical Power Analysis in Deep Reinforcement Learning Experiments.” [Primary manuscript](#). §§4–5 relate seeds, variance, power and effect size; §5.1 uses same-algorithm null comparisons. Supports rejecting a universal seed rule. Does not certify a pooled EE ratio with crossed simulator/training factors.

R28 — Agarwal, R.; Schwarzer, M.; Castro, P. S.; Courville, A.; Bellemare, M. G. (2021), “Deep Reinforcement Learning at the Edge of the Statistical Precipice.” *NeurIPS*. [Primary manuscript](#). §§3–4, Fig. 5 and Appendix A.5 support few-run uncertainty and evaluation-protocol sensitivity. Robust benchmark aggregates are not automatic substitutes for the project's chosen pooled EE estimand.

R29 — Deng, A.; Knoblich, U.; Lu, J. (2018), “Applying the Delta Method in Metric Analytics: A Practical Guide with Novel Ideas.” *KDD*. [Primary paper](#). §§2.2–3.3 support ratio covariance and analysis-unit dependence treatment. The review's paired-log EE influence expression is a derived application, not a formula empirically validated for this simulator.

R30 — Owen, A. B.; Eckles, D. (2012), “Bootstrapping data arrays of arbitrary order.” *Annals of Applied Statistics* 6(3), 895–927. DOI 10.1214/12-AOAS547. **Foundational statistics.** [Primary manuscript](#). Theorems 1–2 address resampling crossed arrays under specified structures. Supports dependence-aware reasoning; neither arbitrary dependence nor exact few-cluster ratio coverage is guaranteed.

R31 — Glasserman, P.; Yao, D. D. (1992), “Some Guidelines and Guarantees for Common Random Numbers.” *Management Science* 38, 884–908. **Foundational simulation statistics.** [Author-institution record](#). The primary abstract supports conditional variance-reduction guarantees. Full proof was not inspected; no unconditional CRN benefit or realistic fading-correlation claim is made.

R32 — Lakens, D. (2017), “Equivalence Tests: A Practical Primer for t Tests, Correlations, and Meta-Analyses.” *Social Psychological and Personality Science*. DOI 10.1177/1948550617697177. [Primary article](#). Supports distinguishing nonsignificance from evidence of negligible effects and explicitly defining equivalence bounds. The one-sided exclusion statement for relative EE is a logical application, not a supplied small-sample simulator guarantee.

Source-package key and final evidence boundary

P00–P21 denote the correspondingly numbered files in the supplied r6.1 ZIP. Most consequential: P01 chronology; P03–P06 old oracle/search results; P11 assumptions register and corrections; P12 training; P14 evaluation; P15 included probe code; P16 adjudication; P18 differential audit; P19 rate-target analysis; P20 controlling amendment; P21 addenda. These are primary records of the team's claims, not independently validated physical or efficacy evidence.

The literature supports a conditional physical opportunity and a stronger prior for coherent joint selection than for an independently decoded residual scalar. It does not verify a successor effect, a necessary positive third-head marginal, the old renewal attribution, actual payload calibration, or statistical exclusion of the requested effect range. Those limits remain explicit rather than being filled with assumed parameter values or invented experiments.